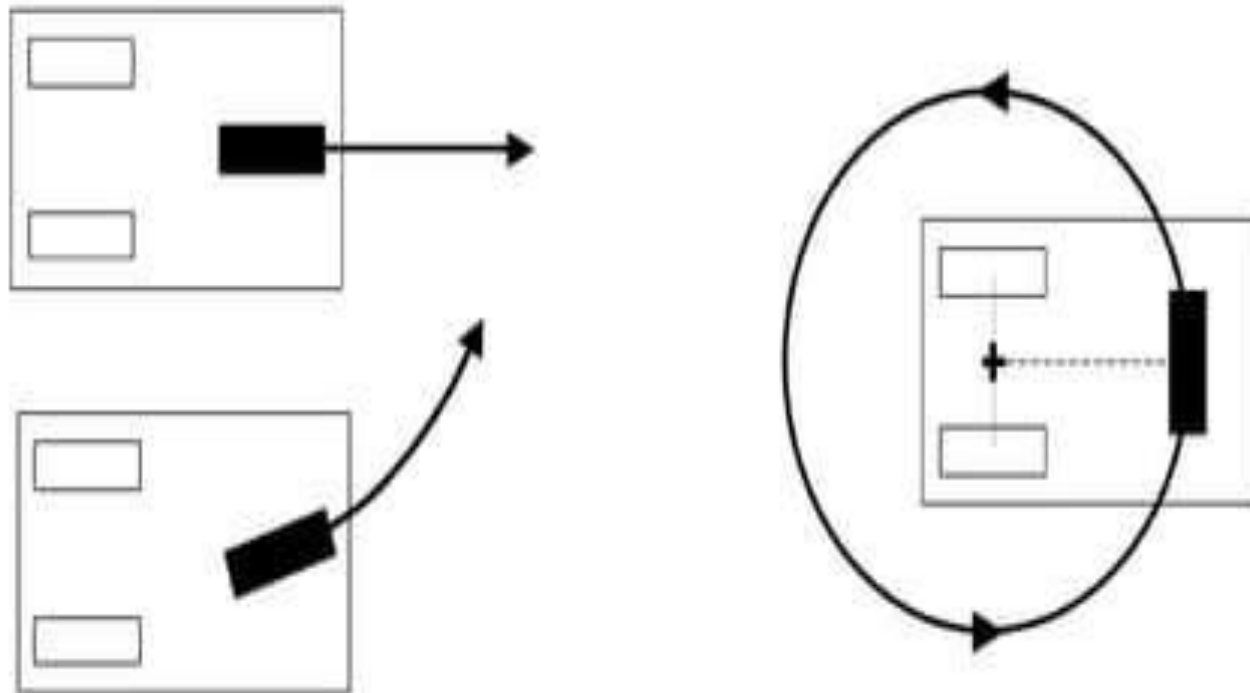


Omnidirectional Wheeled Mobile Robots.

module 3

Single Wheel Drive

- Having a single wheel that is both driven and steered is the simplest conceptual design for a mobile robot.
- Two passive caster wheels in the back.

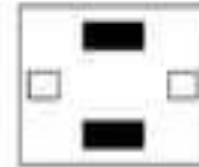
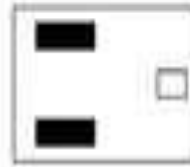


Caster Wheel

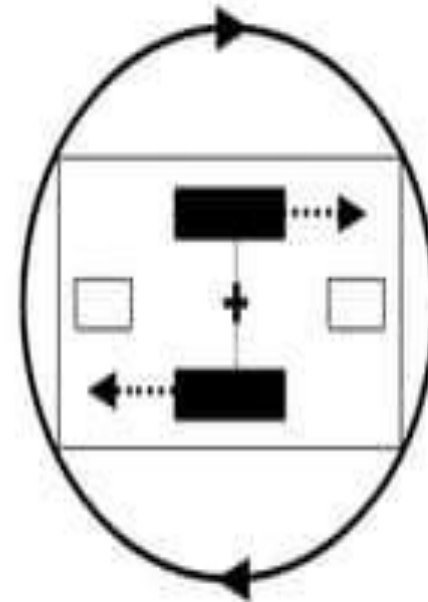
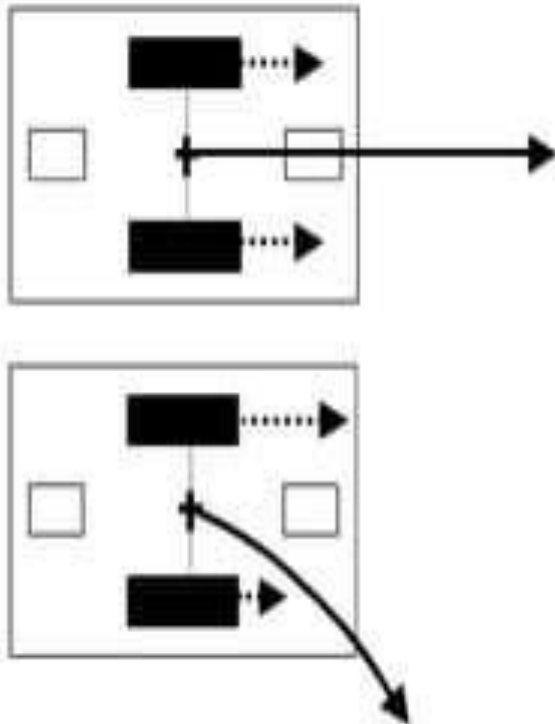


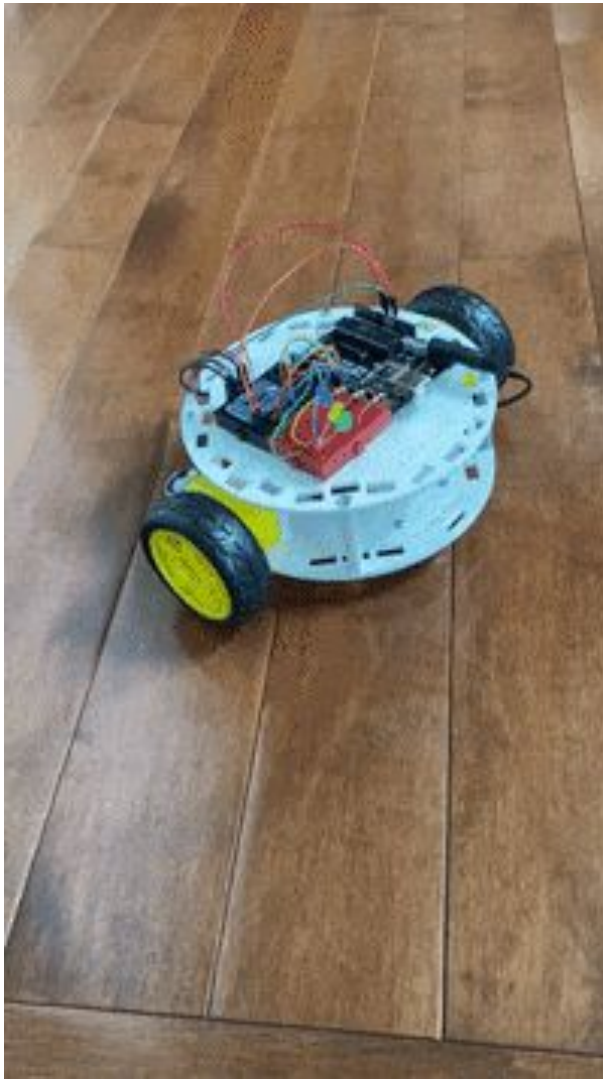
Abin C Jose
Assistant Professor
ADS, MES Kuttippuram

Differential Drive



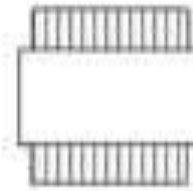
- The differential drive design has two motors mounted in fixed positions on the left and right side of the robot, independently driving one wheel each.



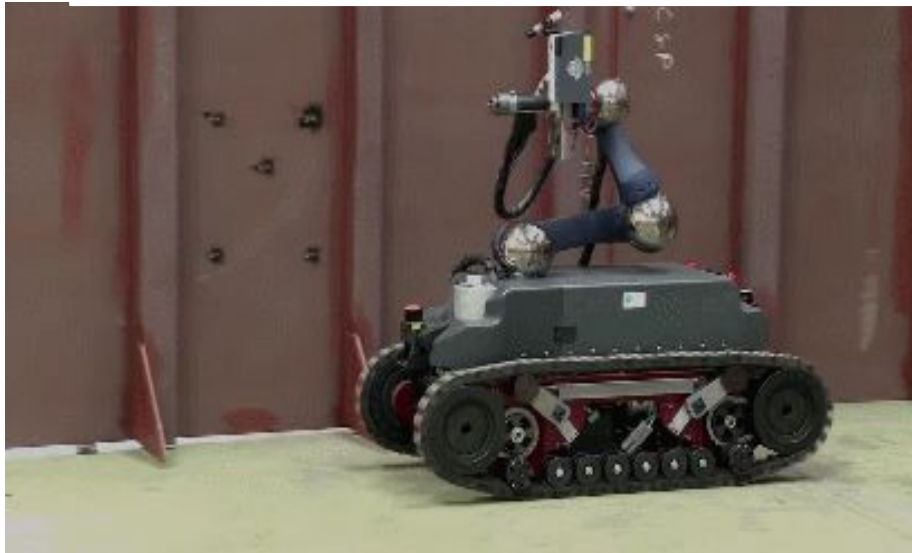


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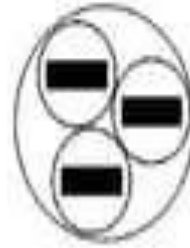
Tracked Robots



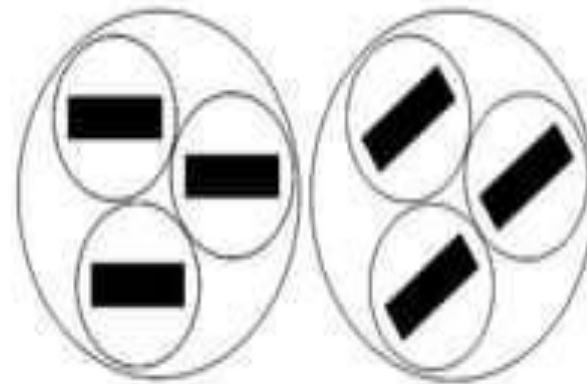
- A tracked mobile robot can be seen as a special case of a wheeled robot with differential drive.



Synchro-Drive



- Synchro-drive is an extension to the robot design with a single driven and steered wheel.
- The three wheels are rotated together so they always point in the same driving direction.



Difference Between Holonomic and Non-Holonomic Robots

Holonomic Robot

- A holonomic robot has the ability to move in any direction instantaneously, meaning the number of controllable degrees of freedom (DOF) is equal to its total degrees of freedom.
- These robots have minimal motion constraints and can achieve omnidirectional movement.

Non Holonomic Robot

- A non-holonomic robot has constraints on its movement, meaning the number of controllable degrees of freedom is less than its total degrees of freedom.
- These robots cannot move in all directions freely and typically require a combination of movements to reach certain positions

Difference Between Holonomic and Non-Holonomic Robots

Holonomic Robot

- **Degrees of Freedom:** The controllable DOF equals the total DOF.
- **Motion Control:** Can move in any direction without requiring rotation.
- **Wheels/Mechanisms Used:** Omnidirectional wheels (Mecanum wheels, Swedish wheels) or spherical wheels.

Examples:

- Omnidirectional mobile robots with Mecanum wheels.
- Autonomous service robots in warehouses.

Non Holonomic Robot

- **Degrees of Freedom:** The controllable DOF is less than the total DOF.
- **Motion Control:** Requires multiple movements (e.g., turning and moving forward) to achieve certain displacements.
- **Wheels/Mechanisms Used:** Differential drive, car-like steering, and Ackermann steering.

Examples:

- Differential drive robots (e.g., TurtleBot).
- Car-like autonomous vehicles
- Robotic arms with constrained joint movements.



Omni-directional robot



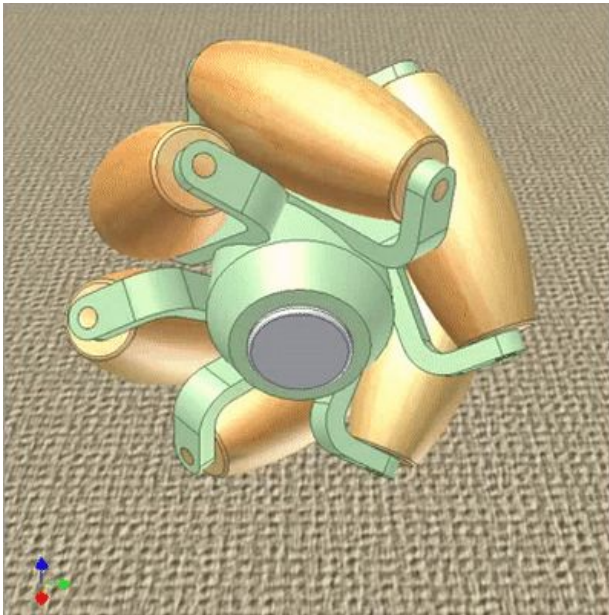
Omni-directional robot

- Above robots cannot drive in all possible directions.
- For this reason, these robots are called “non-holonomic”.
- In contrast, a “holonomic” or omni-directional robot is capable of driving in any direction.
- They are capable of driving in any direction in a 2D plane.

Types of Omnidirectional Wheels

Mecanum Wheel

- The Mecanum wheel is a specialized wheel design that enables omnidirectional movement for vehicles and robots.
- Unlike conventional wheels, Mecanum wheels allow movement in any direction without needing to change orientation.



JAMESPROVOST.COM



Mecanum Wheels

Perfect for tight spaces, they enable



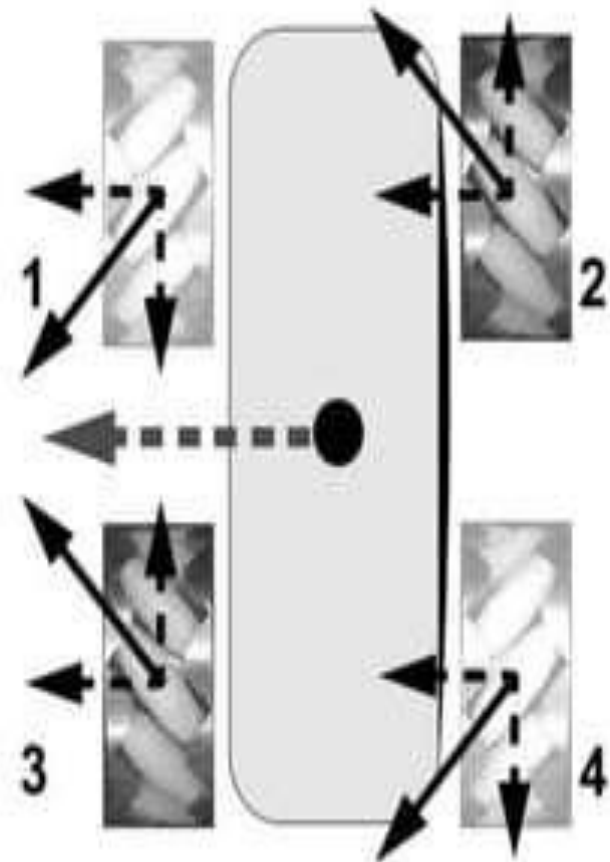
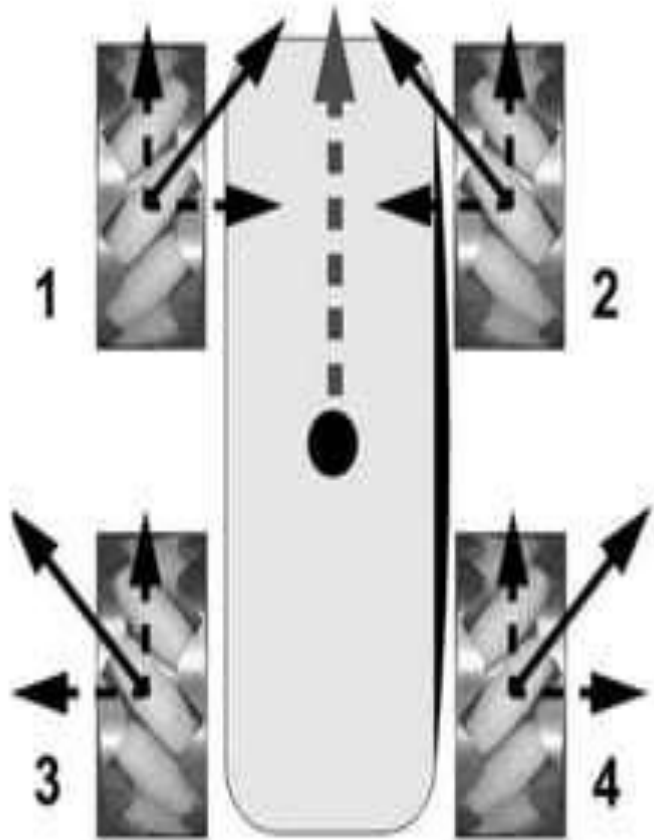
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Omni Wheels (Swedish Wheels)

- Consist of multiple smaller rollers placed perpendicular to the wheel's main axis, enabling smooth lateral movement.

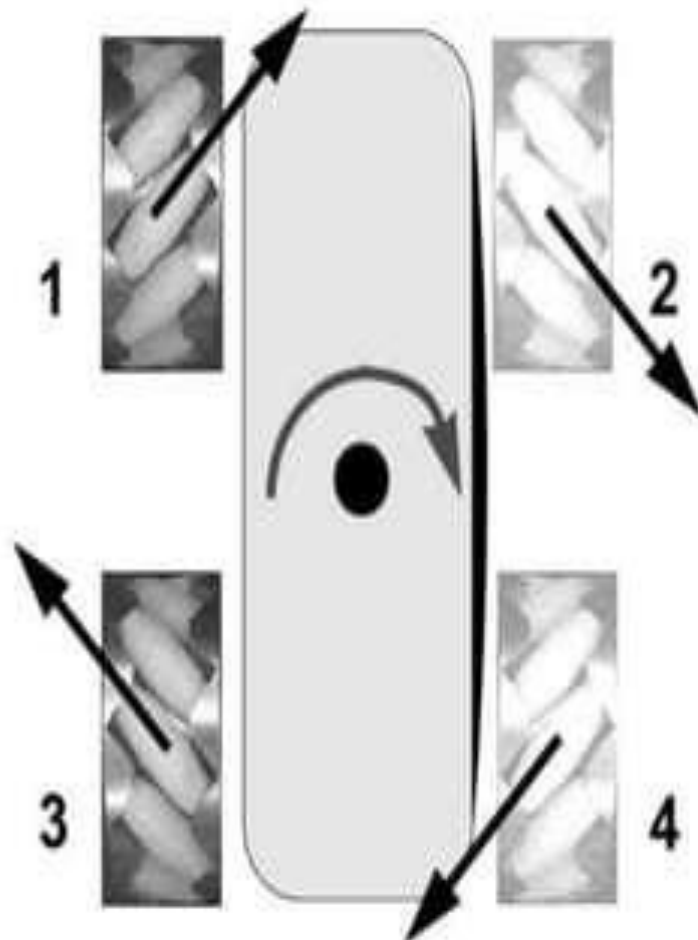


Omni-Directional Drive



Mecanum principle, driving forward and sliding sideways;
dark wheels rotate forward, bright wheels backward (seen from below)

Omni-Directional Drive



Mecanum principle, turning clockwise (seen from below)

Omni-Directional Drive

- Driving forward: all four wheels forward
- Driving backward: all four wheels backward
- Sliding left: 1, 4: backward; 2, 3: forward
- Sliding right: 1, 4: forward; 2, 3: backward
- Turning clockwise on the spot: 1, 3: forward; 2, 4: backward
- Turning counter-clockwise: 1, 3: backward; 2, 4: forward

